



8789 - Disentangle the puzzling jets/outflows in a young protobinary system

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	HOPS-361 NIRSpec	NIRSpec IFU Spectroscopy	(1) HOPS-361-C

ABSTRACT

Jets and outflows facilitate the mass accretion during protostellar evolution and offer a valuable probe for the mass growth of protostars. We have identified a unique young protobinary system (IRS3) that is in a very early stage of disk fragmentation, with both components still actively acquiring mass from spiral arms and streamers. Both protostars are associated with a spatially resolved jet on 100au scale, seen in radio continuum and molecular SiO, respectively. However, neither jet could account for the CO outflow on a few 1000au scale, unless strong recollimation or bending occurs shortly after jet launch. We propose JWST NIRSpec IFU observations to better characterize the jet material in different phases, thereby disentangle the puzzling relationship of the jets/outflows. Given that the system is a textbook example with gas streamers, spiral arms, and binary jets coexisting at a favorable inclination, IRS3 also offers a rare chance to trace the interplay of gas infall, flow, and ejection across different scales and elucidate the evolution of a protobinary system right after disk fragmentation.

OBSERVING DESCRIPTION

This proposal focuses on a distinctive proto-binary system formed via disk fragmentation. We request JWST NIRSpec IFU mode a square field of view of 3"x3" with a continuous spectro-imaging coverage from 0.97 to 5.27 μm . We select the F100LP (1--1.9 μm), F170LP (1.7--3.2 μm), and F290LP (2.9--5.3 μm) filters along with G140H, G235H, and G395H dispersers for the highest spectral resolution ($R \sim 2700$ or $\Delta V = 90\text{--}150$ km/s). This range covers transitions from multiple atomic (HI, HeI, OI, FeI...), molecular (CO, H₂), and ionized ([Fe II]) tracers, and allow for a detailed investigation of different phases/compositions of the binary jets.

Proposal 8789 - Targets - Disentangle the puzzling jets/outflows in a young protobinary system

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	HOPS-361-C	RA: 05 47 4.6337 (86.7693071d) Dec: +00 21 47.80 (.36328d) Equinox: J2000	Epoch of Position: 2000.0	
Comments: Category=Star Description=[Protostars, Young stellar objects] Extended=YES					

Proposal 8789 - Observation 1 - Disentangle the puzzling jets/outflows in a young protobinary system

Observation	Proposal 8789, Observation 1: HOPS-361 NIRSpec												Fri Nov 28 15:00:20 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 1:1) Warning (Form): Data Excess over lower threshold												
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(1)	HOPS-361-C	RA: 05 47 4.6337 (86.7693071d) Dec: +00 21 47.80 (.36328d) Equinox: J2000					Epoch of Position: 2000.0					
	Comments: Category=Star Description=[Protostars, Young stellar objects] Extended=YES												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	CYCLING			MEDIUM		10		6				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G140H/F100LP	NRSIRS2RAPID	20	3	false	true	NONE	6	18	5514.6		
	2	G140H/F100LP	NRSIRS2RAPID	10	3	true	true	NONE	6	18	2888.6		
	3	G235H/F170LP	NRSIRS2RAPID	20	2	false	true	NONE	6	12	3676.4		
	4	G235H/F170LP	NRSIRS2RAPID	10	2	true	true	NONE	6	12	1925.733		
	5	G395M/F290LP	NRSIRS2RAPID	20	1	false	true	NONE	6	6	1838.2		
	6	G395M/F290LP	NRSIRS2RAPID	10	1	true	true	NONE	6	6	962.867		